Canadian Reindeer PrioScout Swab Report

Methodology

All reactions were run for 72 hours at 42°C.

Sentinel Swabs

Swab pellets were resuspended in 500µL of 0.1% SDS/1X PBS. Samples were run by RT-QuIC in 4 replicates. Results were compared against the negative plate controls using a t-test with the alternative hypothesis being that a sample will have a larger value than the negative control. Metrics analyzed were maxpoint ratio (MPR) and maximum slope (MS). The p-values for each metric must have been ≤ 0.05 and at least 50% of replicates must have crossed a threshold of twice the background fluorescence for a sample to have been considered detected. If a sample had at least 50% of its replicates cross the threshold, but was negative for the other two metrics, it was labelled as “suspect”.

Salt block

The salt block was dissolved into a stock solution of 100mg/mL. The stock solution was then serially diluted 10-fold into 10^{-2}–10^{-4} concentrations. The final concentration of the salt block in the reaction was 20ng/µL, 2ng/µL, and 0.2ng/µL. These concentrations were tested both alone and spiked with an equivalent 10^{-3} concentration of positive control lymph node.

Additionally, the salt block was swabbed directly and processed according to the current MNPRO environmental swab extraction SOP. Pellets from the extraction were resuspended in 500µL of SDS/PBS and tested either alone or spiked with positive control as above.

Results of the spiking were analyzed against the RAFs of a serial titration of positive control to determine the relative effect on the reaction kinetics. Non-spiked samples were observed for any spurious seeding activity.

Swab Results

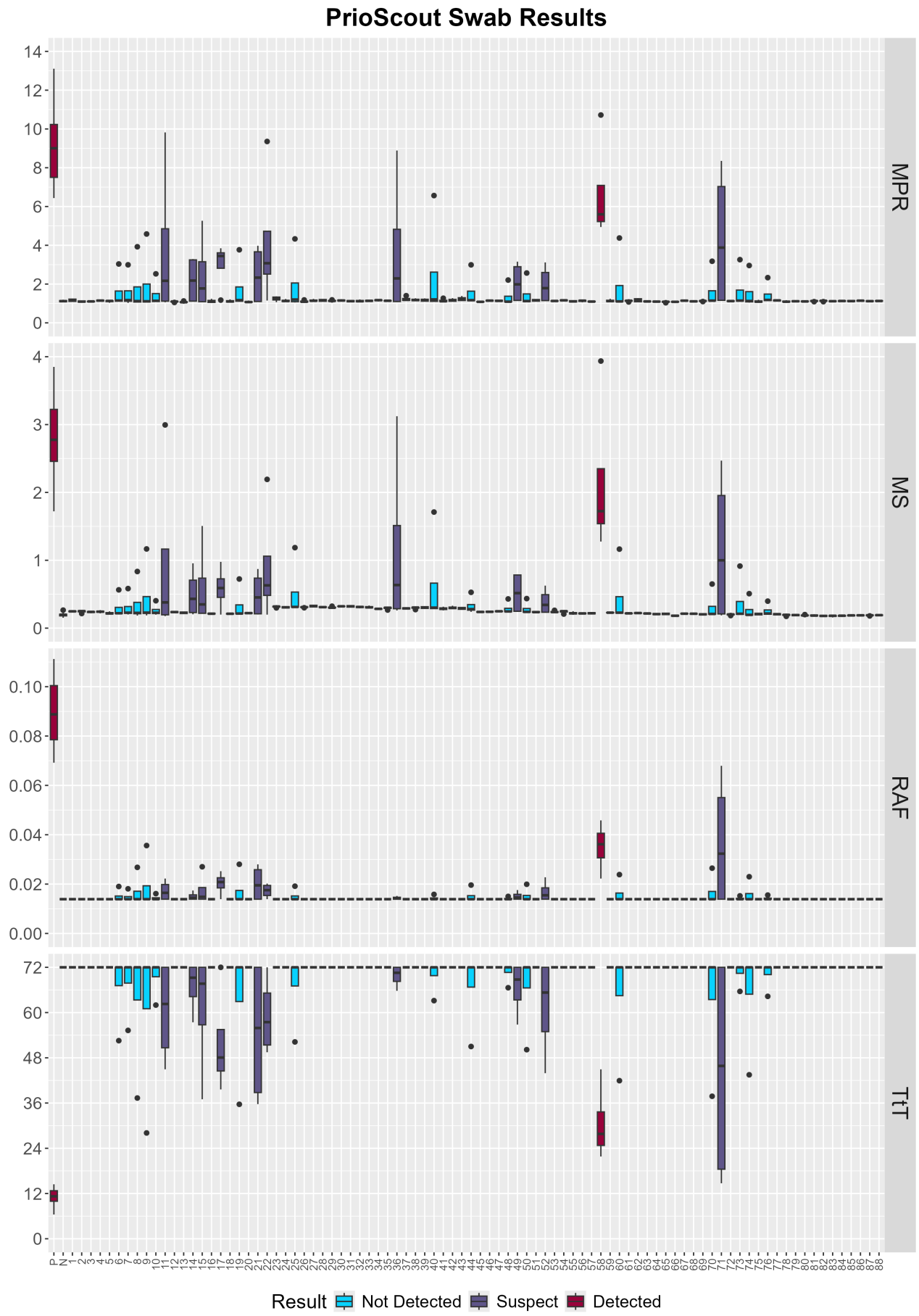
- Positives: **58**
- Suspects: **11, 14, 15, 17, 21, 22, 36, 49, 52, 71**

Table 1: Results of RT-QuIC testing. Samples were given a result of 'Not Detected' or 'Detected' if it was positive for zero or all three metrics, respectively. If a sample was threshold positive (i.e. more than half of replicates crossed threshold) but was negative for either of the other two metrics, it was designated as 'Suspect'. crossed: fraction of replicates above threshold; thres pos: at least half of replicates crossed threshold?; MPR: maxpoint ratio; MS: max slope.

Sample IDs	crossed	thres pos	MPR pvalue	MS pvalue	Result
1	0	FALSE	0.25	0.162	Not Detected
2	0	FALSE	0.608	0.174	Not Detected
3	0	FALSE	0.661	0.211	Not Detected
4	0	FALSE	0.35	0.183	Not Detected
5	0	FALSE	0.569	0.295	Not Detected
6	0.25	FALSE	0.163	0.157	Not Detected
7	0.25	FALSE	0.165	0.149	Not Detected
8	0.25	FALSE	0.171	0.157	Not Detected
9	0.25	FALSE	0.171	0.155	Not Detected
10	0.25	FALSE	0.166	0.166	Not Detected
11	0.5	TRUE	0.12	0.139	Suspect
12	0	FALSE	0.695	0.205	Not Detected
13	0	FALSE	0.759	0.271	Not Detected
14	0.5	TRUE	0.0827	0.0848	Suspect
15	0.5	TRUE	0.112	0.117	Suspect
16	0	FALSE	0.519	0.346	Not Detected
17	0.75	TRUE	0.041	0.0543	Suspect
18	0	FALSE	0.526	0.372	Not Detected
19	0.25	FALSE	0.16	0.157	Not Detected
20	0	FALSE	0.725	0.296	Not Detected
21	0.5	TRUE	0.0843	0.0839	Suspect
22	0.75	TRUE	0.0834	0.0889	Suspect
23	0	FALSE	0.113	0.0511	Not Detected
24	0	FALSE	0.375	0.0557	Not Detected
25	0.25	FALSE	0.159	0.0996	Not Detected
26	0	FALSE	0.58	0.054	Not Detected
27	0	FALSE	0.276	0.0469	Not Detected
28	0	FALSE	0.288	0.0533	Not Detected
29	0	FALSE	0.374	0.0563	Not Detected
30	0	FALSE	0.283	0.0479	Not Detected
31	0	FALSE	0.49	0.0482	Not Detected
32	0	FALSE	0.43	0.0537	Not Detected
33	0	FALSE	0.403	0.0547	Not Detected
34	0	FALSE	0.27	0.0776	Not Detected

35	0	FALSE	0.406	0.0658	Not Detected
36	0.5	TRUE	0.112	0.105	Suspect
37	0	FALSE	0.119	0.0696	Not Detected
38	0	FALSE	0.169	0.0631	Not Detected
39	0	FALSE	0.219	0.0591	Not Detected
40	0.25	FALSE	0.159	0.123	Not Detected
41	0	FALSE	0.333	0.0748	Not Detected
42	0	FALSE	0.231	0.0613	Not Detected
43	0	FALSE	0.111	0.0688	Not Detected
44	0.25	FALSE	0.155	0.0857	Not Detected
45	0	FALSE	0.712	0.196	Not Detected
46	0	FALSE	0.358	0.174	Not Detected
47	0	FALSE	0.366	0.159	Not Detected
48	0.25	FALSE	0.191	0.111	Not Detected
49	0.5	TRUE	0.0796	0.0686	Suspect
50	0.25	FALSE	0.176	0.12	Not Detected
51	0	FALSE	0.27	0.191	Not Detected
52	0.5	TRUE	0.0844	0.0788	Suspect
53	0	FALSE	0.442	0.19	Not Detected
54	0	FALSE	0.261	0.189	Not Detected
55	0	FALSE	0.552	0.308	Not Detected
56	0	FALSE	0.463	0.332	Not Detected
57	0	FALSE	0.612	0.312	Not Detected
58	1	TRUE	0.0315	0.0382	Detected
59	0	FALSE	0.383	0.254	Not Detected
60	0.25	FALSE	0.171	0.15	Not Detected
61	0	FALSE	0.439	0.327	Not Detected
62	0	FALSE	0.254	0.292	Not Detected
63	0	FALSE	0.588	0.341	Not Detected
64	0	FALSE	0.663	0.399	Not Detected
65	0	FALSE	0.766	0.376	Not Detected
66	0	FALSE	0.723	0.613	Not Detected
67	0	FALSE	0.335	0.353	Not Detected
68	0	FALSE	0.559	0.378	Not Detected
69	0	FALSE	0.491	0.437	Not Detected
70	0.25	FALSE	0.158	0.164	Not Detected
71	0.5	TRUE	0.0823	0.0845	Suspect
72	0	FALSE	0.475	0.478	Not Detected
73	0.25	FALSE	0.16	0.159	Not Detected
74	0.25	FALSE	0.175	0.207	Not Detected
75	0	FALSE	0.572	0.41	Not Detected
76	0.25	FALSE	0.146	0.186	Not Detected
77	0	FALSE	0.317	0.434	Not Detected
78	0	FALSE	0.591	0.535	Not Detected
79	0	FALSE	0.552	0.545	Not Detected
80	0	FALSE	0.566	0.564	Not Detected
81	0	FALSE	0.462	0.58	Not Detected

82	0	FALSE	0.452	0.635	Not Detected
83	0	FALSE	0.542	0.649	Not Detected
84	0	FALSE	0.498	0.651	Not Detected
85	0	FALSE	0.421	0.6	Not Detected
86	0	FALSE	0.418	0.579	Not Detected
87	0	FALSE	0.502	0.574	Not Detected
88	0	FALSE	0.415	0.565	Not Detected


PrioScout Swab Results
MPR
MS
RAF
TtT
Result
Not Detected
Suspect
Detected

Salt Block Results

The dissolved salt solutions exhibited no false seeding activity within the 72 hour time frame; however, when the blocks were swabbed and processed, the insoluble pellet showed significant seeding activity around the 50 hour mark.

Interestingly, each method of spiking the salt concentrations showed an relative increase in the speed of the reaction with the insoluble pellet from the swab protocol essentially doubling the speed of the reaction.

This study concludes that the chosen salt block has a significant effect on the kinetics of the reaction. Although, at the concentrations expected from swabbing a sentinel and not the block directly, we likely would not see false seeding activity in the given time frame.

Table 2: Summary table of the salt block results. Kinetics were compared using time-to-threshold (TtT) and its inverse, rate of amyloid formation (RAF). Dilutions are represented in log form. pos_dilution: dilution factor of positive control; block_dilution: dilution factor of salt block; N: negative control; P: positive control; B: salt block; BP: salt block spiked with positive control; S: salt block swab; SP: salt block swab spiked with positive control.

Sample IDs	pos_dilution	block_dilution	median_TtT	median_RAF
N	NA	NA	72.000	0.01389
P	-3	NA	11.460	0.08724
BP	-3	-2	9.158	0.11000
BP	-3	-3	9.490	0.10550
BP	-3	-4	10.400	0.09623
B	NA	-2	72.000	0.01389
B	NA	-3	72.000	0.01389
B	NA	-4	72.000	0.01389
SP	-3	NA	5.815	0.17230
S	NA	NA	63.360	0.01608

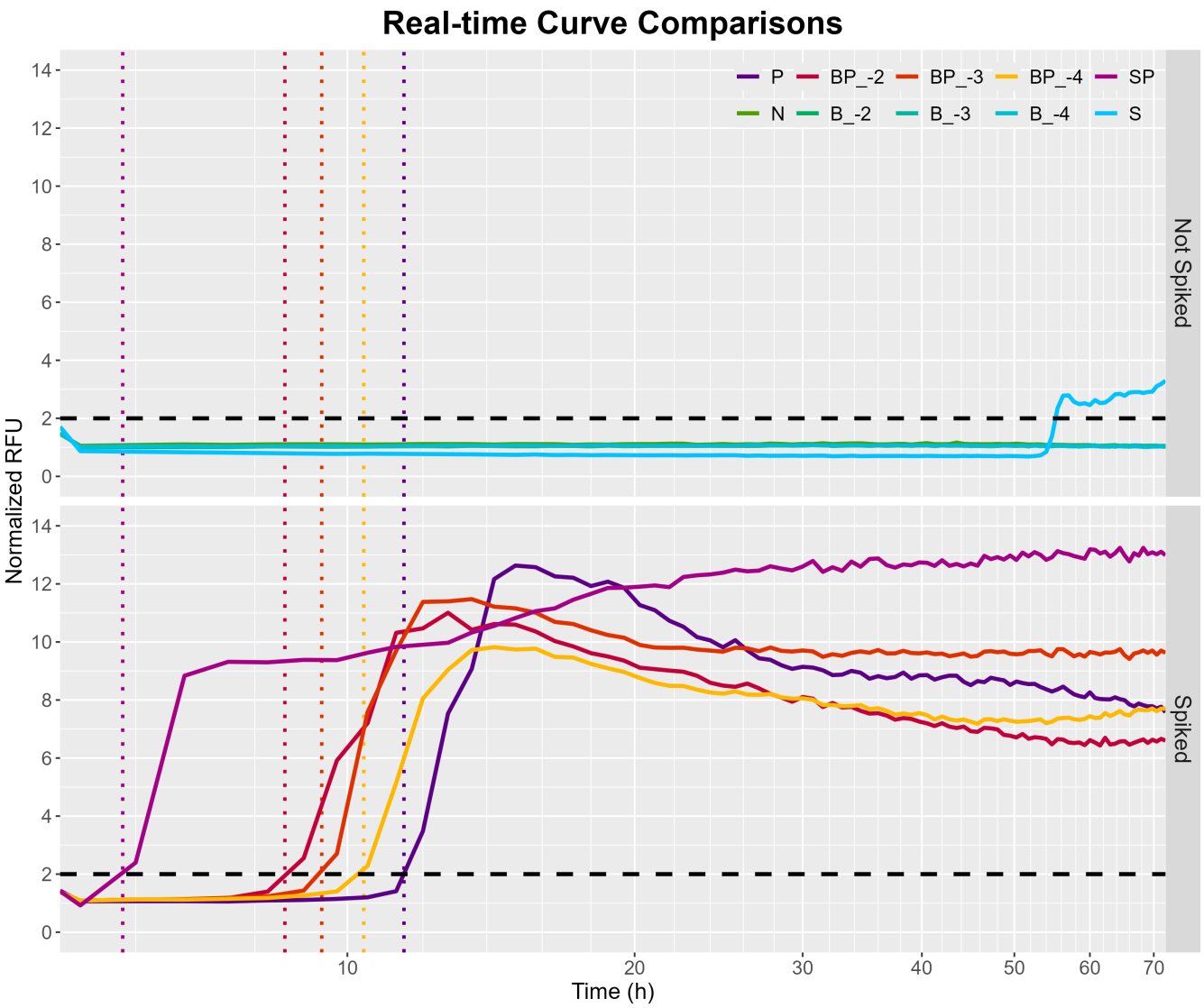
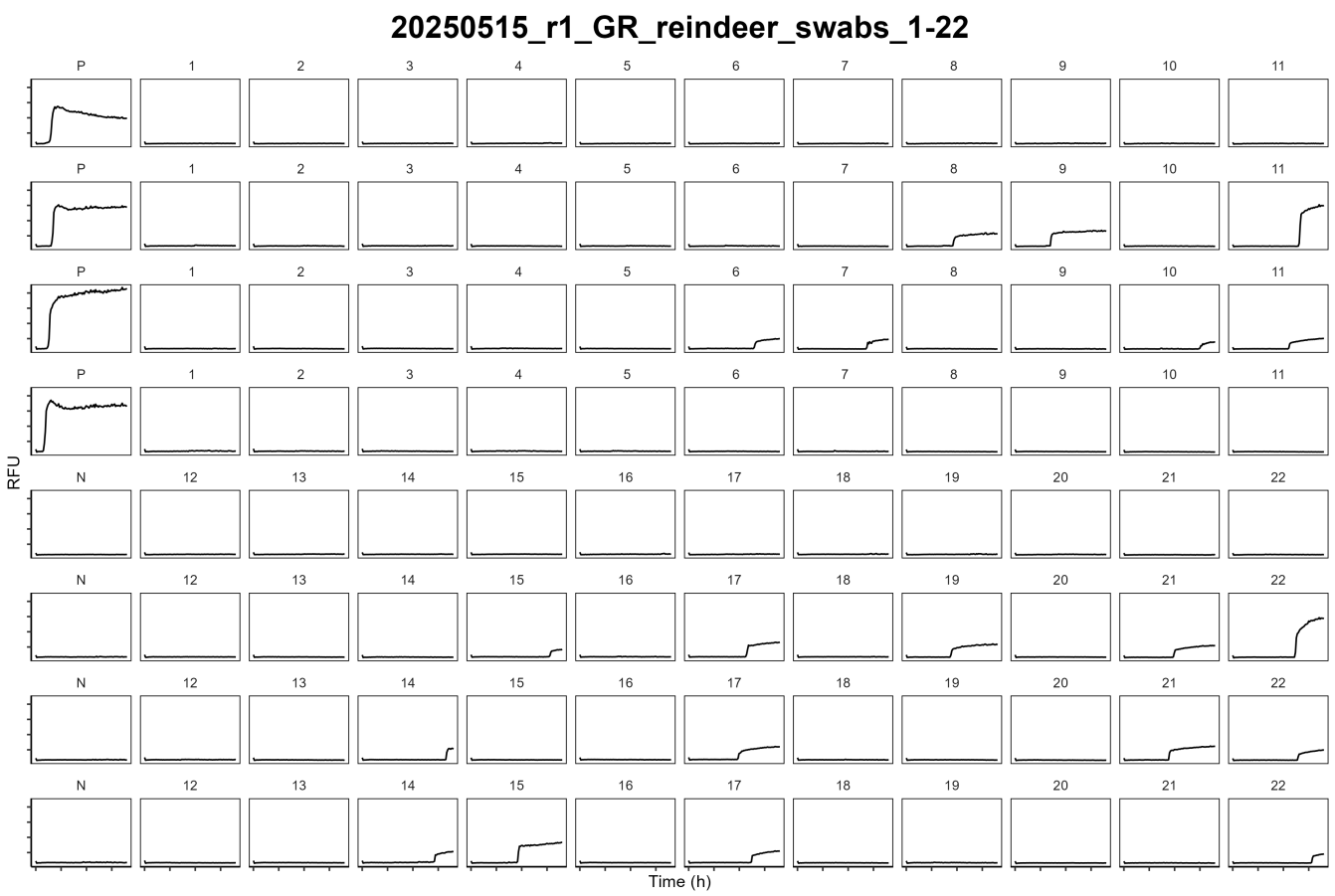
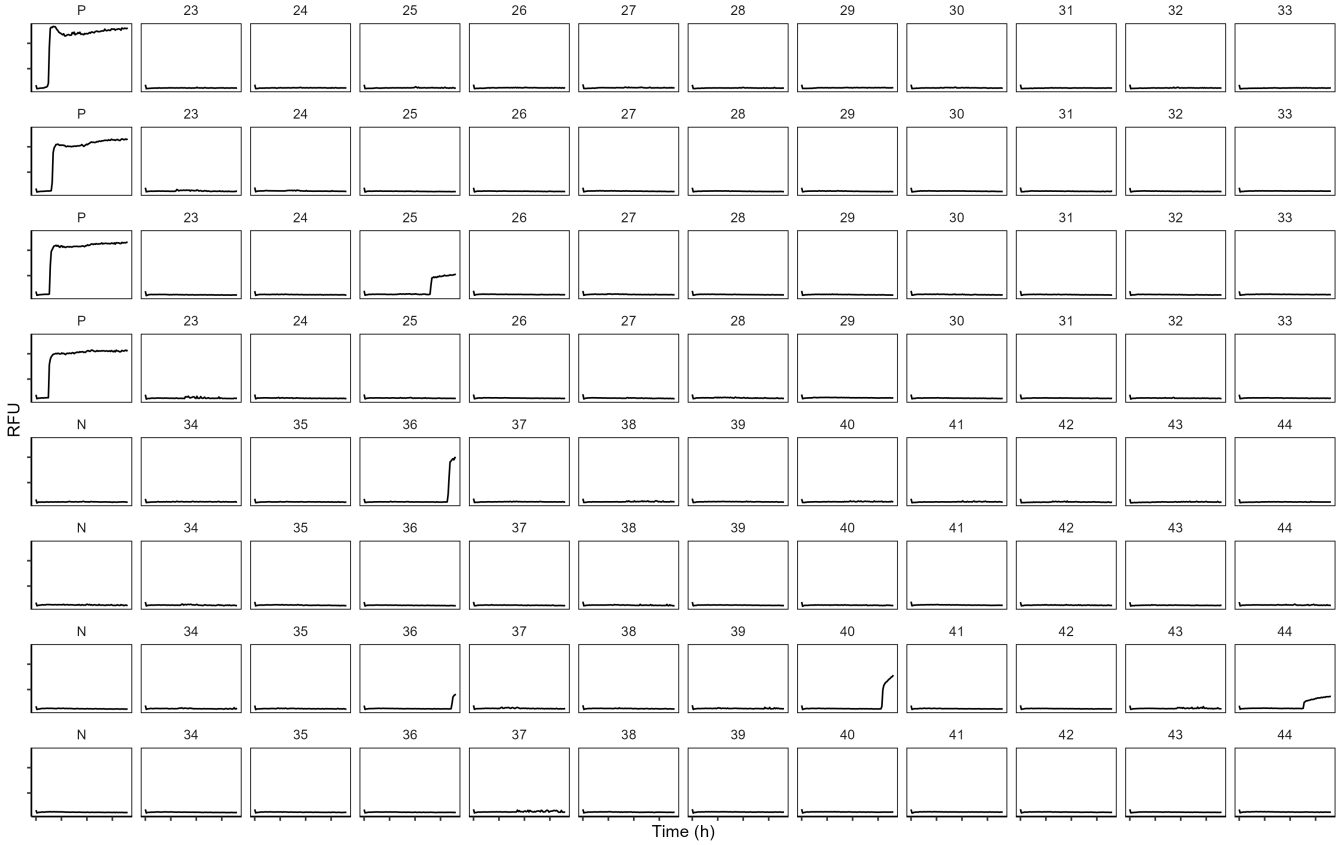


Figure 1: Real-time curves of median normalized fluorescent values. Curves in the top graph represent non-spiked samples; spiked samples are below. For sample key, see Table 2.

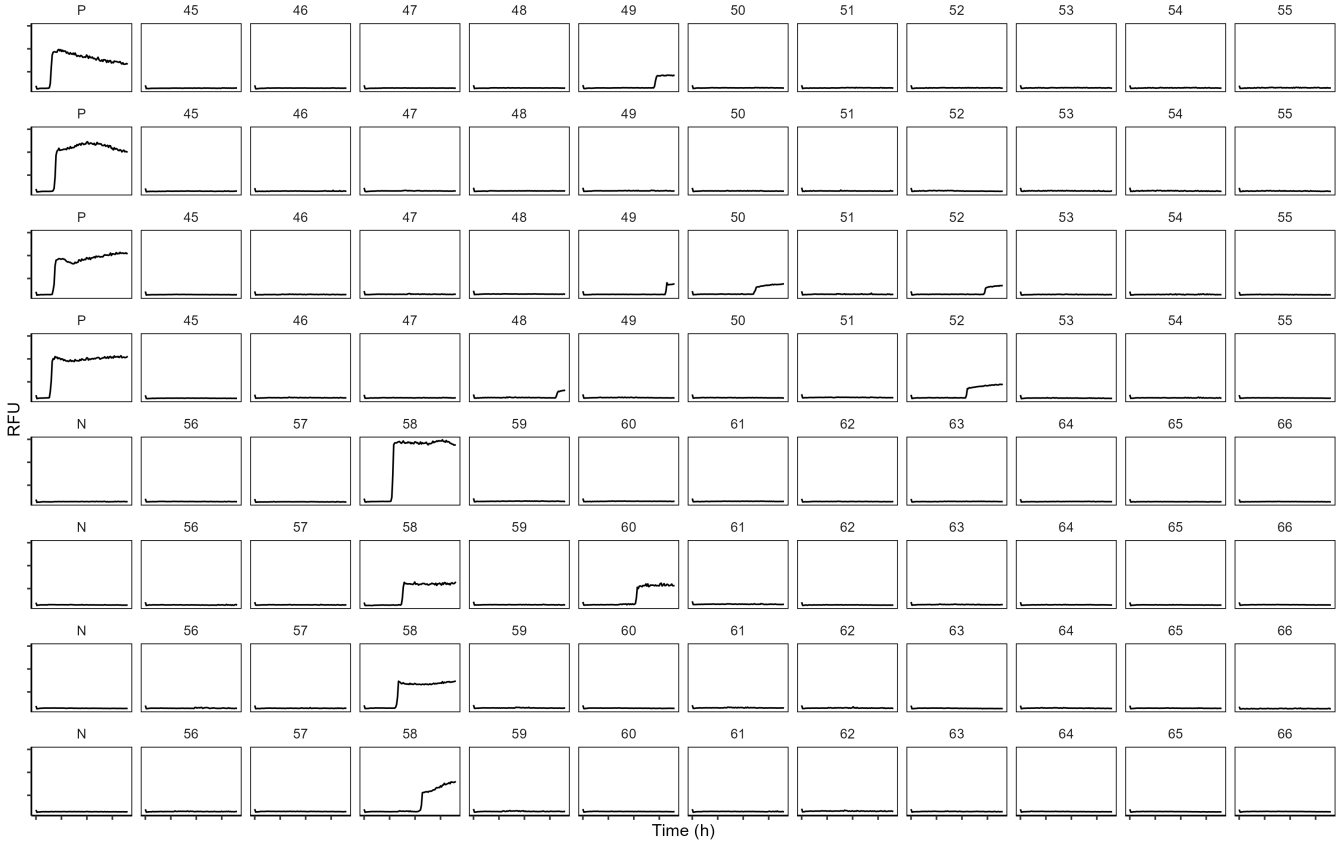
Plate Views


20250515_r1_GR_reindeer_swabs_1-22
P 1 2 3 4 5 6 7 8 9 10 11
N 12 13 14 15 16 17 18 19 20 21 22
RFU
Time (h)

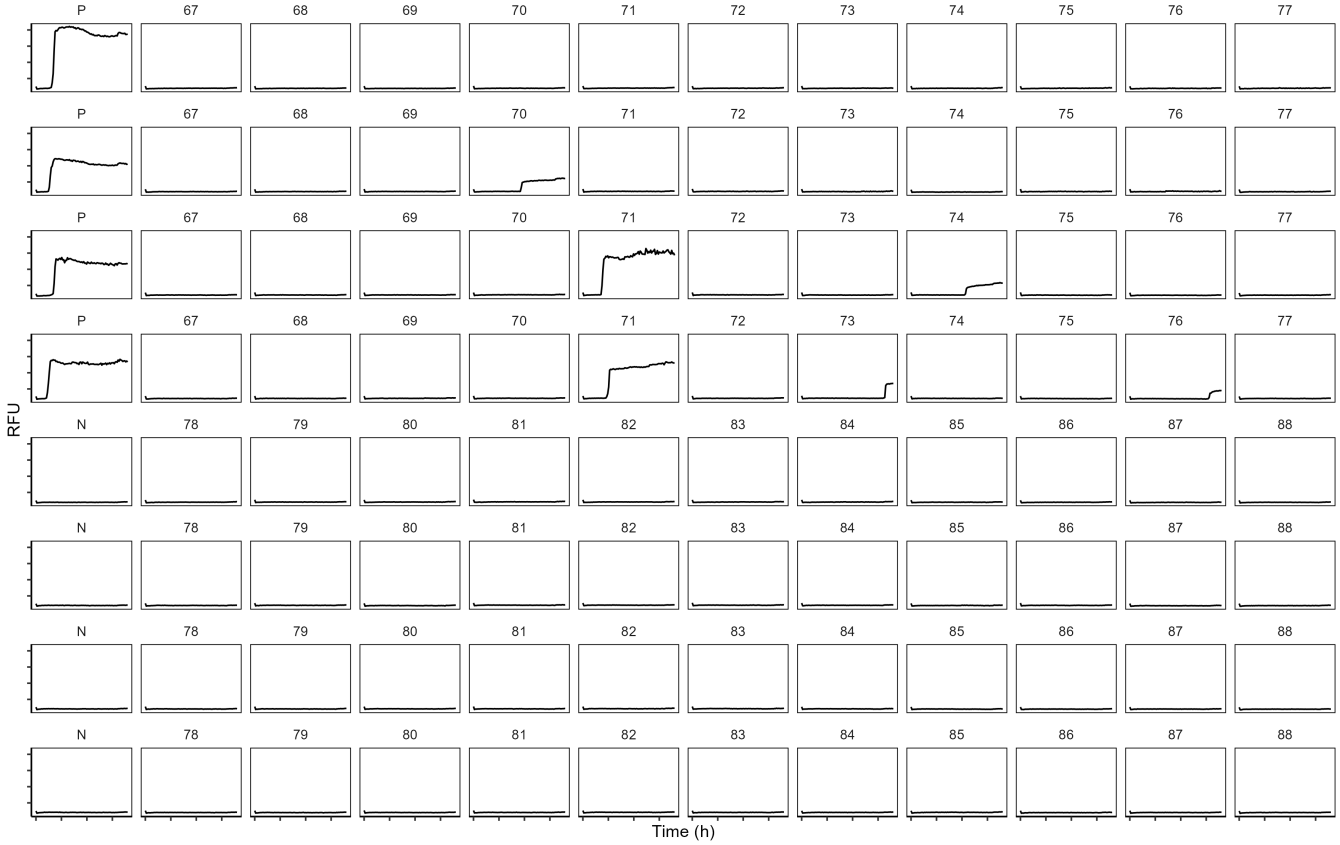
20250515_r2_GR_reindeer_swabs_23-44



20250515_r3_GR_reindeer_swabs_45-66



20250515_r4_GR_reindeer_swabs_67-88



20250523_r6_GR_cobalt_block

